Feasible regions

1 Describe what a feasible region represents with regards to a system of inequalities.

2 Determine which of the following ordered pairs satisfies the system of inequalities:

$$2x - y > -6 \text{ and } 2x + 2y < 0$$

A $(2, 6)$ **B** $(-3, 6)$ **C** $(1, -3)$ **D** $(-6, 1)$

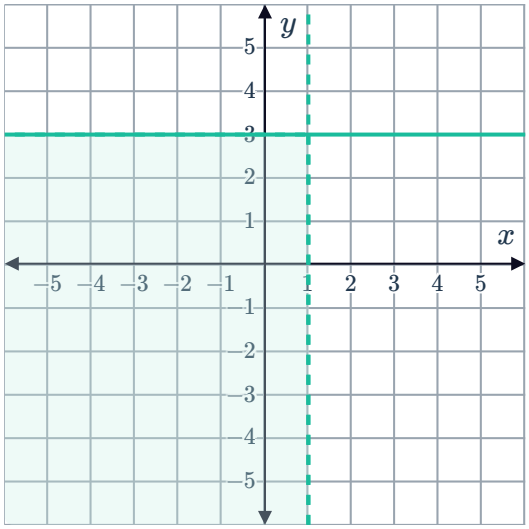
3 Without graphing them, determine whether the following systems of inequalities have no solutions or infinitely many solutions. Explain your answer.

a $9x - y < 7$ and $9x - y > 7$

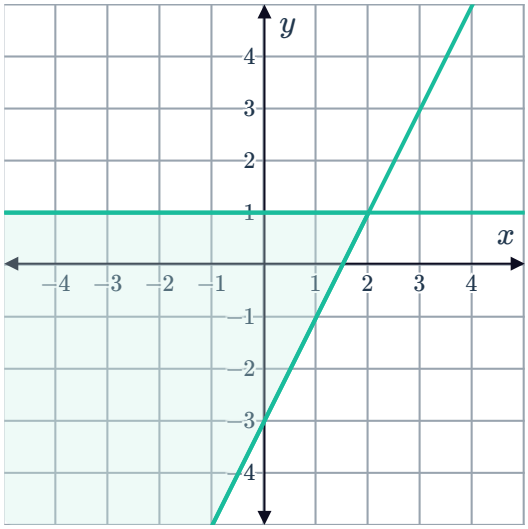
b $2x - y \leq 7$ and $2x - y \geq 7$

4 For each of the following graphs, state the system of inequalities that describe the shaded region:

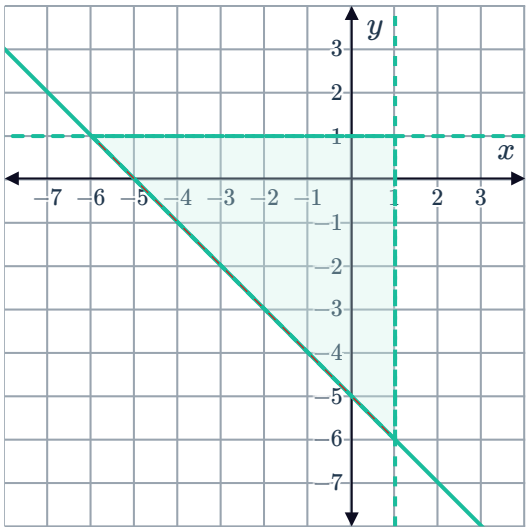
a



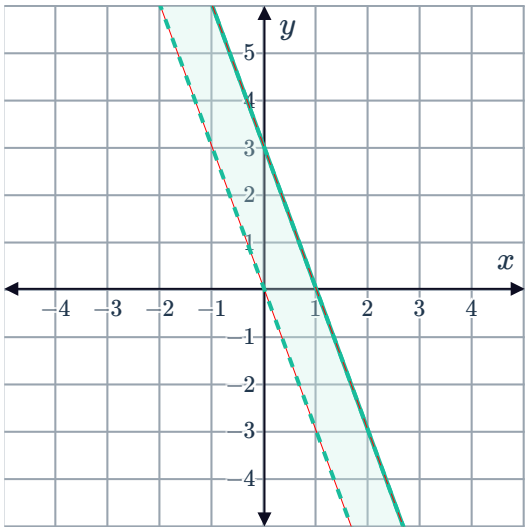
b



c



d



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5 Consider the system of inequalities $2x - y < 4$ and $6x + 3y > 0$.

a Sketch the equations $2x - y = 4$ and $6x + 3y = 0$ on the same set of axes.

b Determine whether the point $(2, 2)$ satisfies the system of inequalities.

c Hence graph the solution set to the system of inequalities $2x - y < 4$ and $6x + 3y > 0$.

6 Consider the system of inequalities, $y > x - 5$, $y > -x - 5$, and $y < 0$.

a Graph the lines $y = x - 5$, $y = -x - 5$, and $y = 0$ on the same set of axes.

b Determine whether the point $(1, -2)$ satisfies the system of inequalities.

c Hence graph the region described by the inequalities $y > x - 5$, $y > -x - 5$, and $y < 0$ on the same set of axes.

d Calculate the area of the feasible region.

7 Graph the solution set to each of the following systems of inequalities:

a $x \leq 5$ and $y < 3$

b $y \leq 3x - 4$ and $y > -4x + 1$

c $3x + y > 5$ and $3x + y < 7$

d $y < 4x - 2$ and $y < 3x - 3$

8 Consider the inequalities $x > -2$, $y > 1$ and $y \leq -0.75x + 2.5$.

a Graph the region described by the intersection of the inequalities.

b Calculate the area of the feasible region.

Applications

9 There are x surfers competing for wildcard entry into a pro-surfing competition, of which Joanne is one. She needs her total score from all y judges to be at least 35 in order to be chosen. The total score possible from all judges is at most 50.

a $x > 1$ is one of the inequalities that can be formed for the above situation. Write down the two other inequalities related to y to complete the system.

b Graph the solution to the system of inequalities.

c If Joanne receives scores of $8.4, 7.3, 7.3, 7.8, 8.9$ from the judges, will she be chosen as a wildcard entry?

d Whether Joanne is chosen as a wildcard entry is dependent on the number of surfers competing for the spot. True or false?

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10 Roald has 24 centimetres of leftover wood that he is trying to make a rectangular photo frame out of. Let x represent the length of the photo frame and let y represent the width.

a Write two inequalities to represent the situation.

b Graph solution to the system of inequalities.

c State whether the following could be the dimensions of the photo frame:

i A length of 6 cm and a width of 4 cm

ii A length of 7 cm and a width of 6 cm

iii A length of 9 cm and a width of 8 cm

iv A length of 7 cm and a width of 4 cm

d If Roald wanted to create a square frame, state the dimensions of the largest possible square frame.

11 Tracy has experimented with growing two types of banana trees. She has found that the two types grow best if Type A takes up no more than 35% of the plantation area and Type B takes up at least 20% of the plantation area.

Let x and y represent the decimal proportion of area Type A and Type B banana trees can use respectively.

a Write three inequalities representing the possible values of x and y.

b Graph the solution to the system of the inequalities.

c Use the graph to determine whether the following points correspond to appropriate areas for Type A and Type B banana trees:

i $(0.28, 0.68)$ **ii** $(0.58, 0.68)$ **iii** $(0.13, 0.57)$ **iv** $(0.09, 0.11)$

d Is it possible for Type B trees to grow optimally if they take up 100% of the plantation area? Explain your answer.

12 An airline is checking passengers into two flights, A and B, simultaneously. Due to passenger numbers, there must be at least 10 staff at check-in for flight A and at least 7 staff at check-in for flight B. Since there must be staff on hand for other services, the airline can only allocate at most 23 staff for check-in of both flights.

Let x and y represent the number of staff attending check-in of flights A and B respectively.

a State the system of three inequalities representing the possible values of x and y.

b Graph the solution to the system of inequalities.

c If 14 staff are allocated to checking in passengers of flight A, what is the maximum number of staff that can be allocated to checking in passengers of flight B?

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13 Applicants for a particular university are asked to sit a numeracy test and verbal reasoning test. Successful applicants must obtain a minimum score of 17 on the numeracy test and a minimum combined score of 37 for both tests.

Let x and y represent an applicant's score on the numeracy and verbal reasoning test respectively.

a State the system of three inequalities representing the possible values of x and y.

b Graph the solution to the system of inequalities.

c Determine whether the following points represent scores that would make the applicant successful:

i $(14, 16)$ **ii** $(33, 18)$ **iii** $(20, 13)$ **iv** $(17, 22)$

d If an applicant obtains a score of 24 in the numeracy test, find the minimum integer score they need to obtain in the verbal reasoning test to be successful.

14 Throughout university, Tom works as a mentor, earning $\$15$ per hour, and as a fitness instructor earning $\$18$ per hour. The number of hours he works in each job can vary from week to week, but he never works more than 29 hours in total each week, and he needs to be able to at least cover his weekly expenses of $\$270$.

Let x and y represent the number of hours he works as a mentor and fitness instructor respectively.

a State the system of four inequalities representing the possible values of x and y.

b Graph the solution to the system of inequalities.

c If Tom works 6 hours as a mentor in one week, find the minimum number of hours y he can work as a fitness instructor so that he can cover his expenses.

d Is it possible for him to work the same number of hours in both jobs and still be able to cover his expenses?

15 An airforce plane used for dropping air personnel into enemy territory can fit no more than 32 airmen at a time, and has a maximum carrying capacity of 3900 kilograms. When carrying a parachute and ammunition, an average female's weight is 100 kilograms and an average male's weight is 150 kilograms.

Let x represent the number of women.

Let y represent the number of men.

a Write down the system of inequalities in x and y that represent the number of men and women who can be on the plane at one time. Make y the subject in each inequality.

b Use the graphing application of your CAS calculator to graph the inequalities on the same axis.

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c Determine which of the following combinations of men and women can be on the plane at the same time:

A 21 women and 11 men **B** 19 women and 12 men **C** 25 women and 3 men

D 7 women and 10 men

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